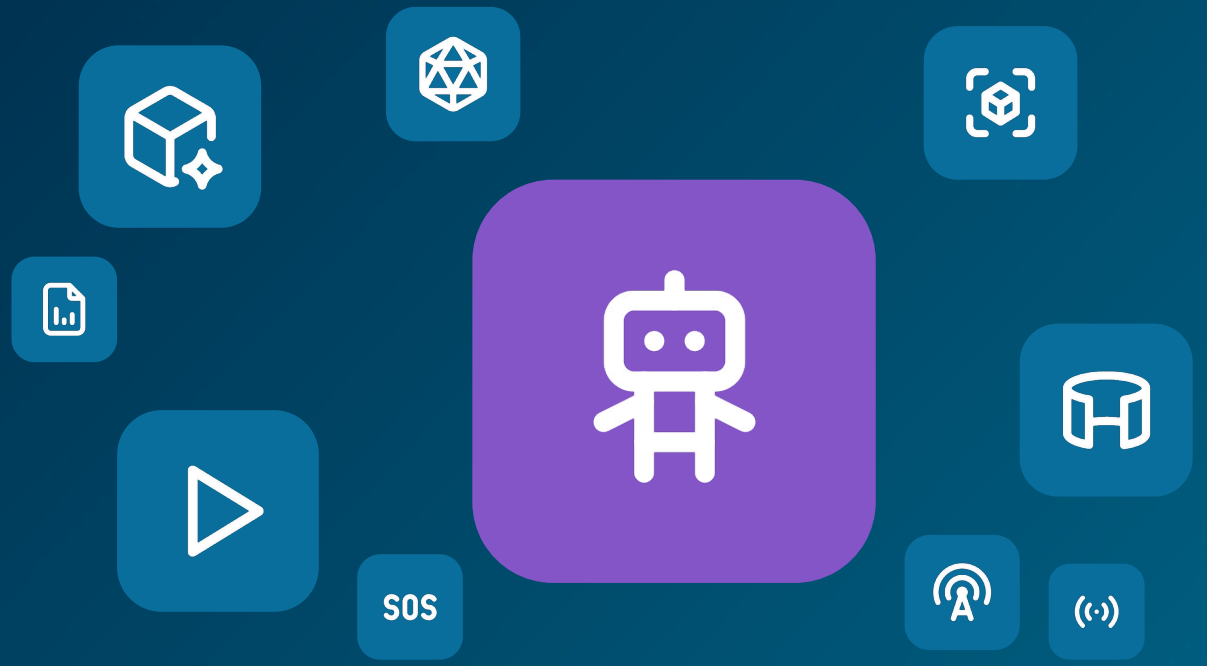
The Media Connectivity Association



OVERVIEW AND UPDATES · IMMERSIVE MEDIA

Conversational Avatar Communication

Real-time conversational avatars, animated from user to user

MPEG (ISO/IEC) 3GPP

OCTOBER 2026 – www.5g-mag.com

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About the project

Real-time conversational avatars, animated from user to user

3GPP TR 26.813

Technical Report

Formats for representing and animating avatars, and recommendations for integrating them in 3GPP services.

MPEG ARF

ISO/IEC 23090-39

The Avatar Representation Format: base avatar data model, container and animation stream.

3GPP TS 26.264

Technical Specification

IMS-based AR real-time communication: avatar calls over the IMS data channel and the three avatar animation modes.

FIND OUT MORE **Technical analysis** **Standards** **Reference Tools**

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The avatar format: MPEG ARF

ISO/IEC 23090-39, as described in 3GPP TR 26.813

Container ISOBMFF or Zip

- ARF document
- Animation track (optional)

Base Avatar Format stores the avatar asset

- Skeletons
- Meshes
- Blendshapes
- Skins
- Landmarks
- Textures
- Nodes (joints)

Animation Stream Format live or stored

- Avatar Animation Units
- AAU header and payload
- configuration
- blendshape
- joint
- landmark
- texture

The base avatar and the animation are separate: a client loads the model once, then receives only the compact animation stream during the call.

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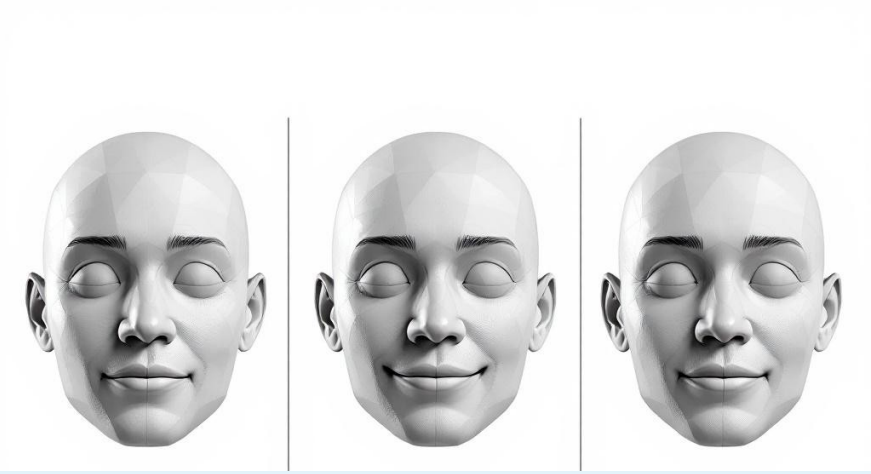
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Animating the avatar

Avatar Animation Units (AAU) of 3GPP TR 26.813

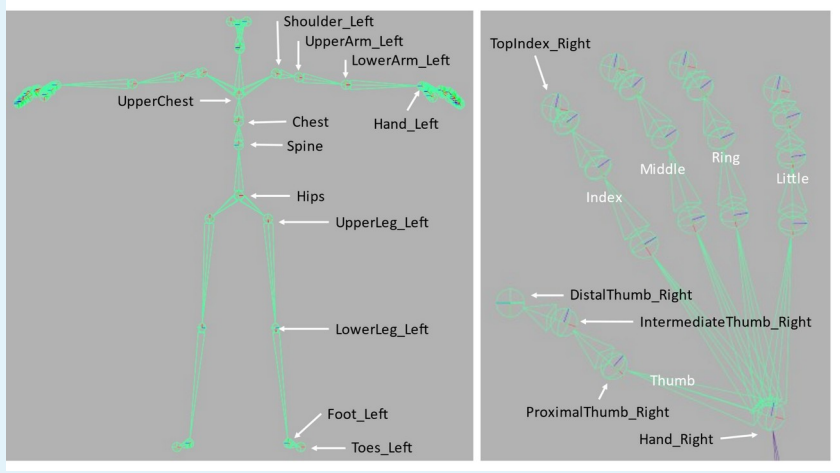
FIVE KINDS OF AAU

configuration | blendshape | joint | landmark | texture



Facial animation

Blend shape animation starts from the neutral expression and adds weighted contributions from predefined expressions.



Body and hand animation

Linear blend skinning moves the avatar surface with the skeleton, weighted per joint.

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Three avatar animation modes

Three avatar animation modes defined by 3GPP TS 26.264 for a call over the IMS data channel

SENDER-CENTRIC

The sender animates

"the sender UE animates and renders its avatar representation before sending it to the receiving UE as 2D video."

RECEIVER-CENTRIC

The receiver animates

"the receiving UE animates and renders the sender UE's avatar representation."

5G-MAG Reference Tools

NETWORK-CENTRIC

The network animates

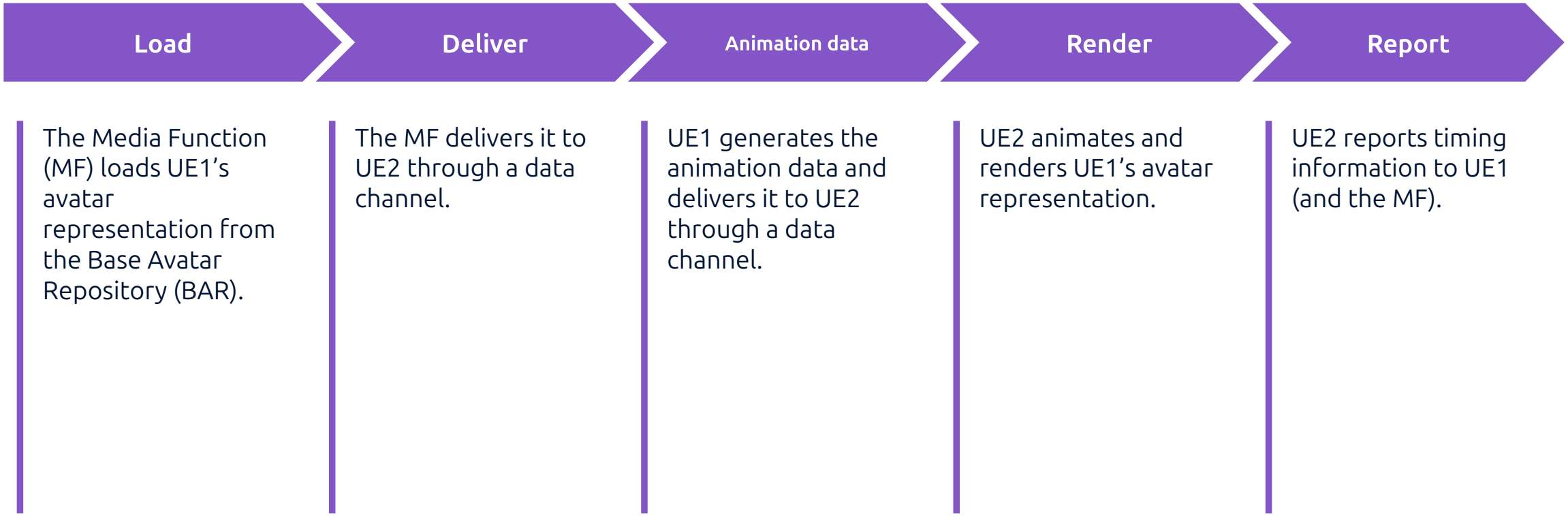
"the MF animates and renders the sender UE's avatar representation before sending it to the receiving UE as 2D video."

Base Avatar Repository: "stores the base avatar of a user and makes it available to other participants" (3GPP TR 26.813).

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A receiver-centric call

Avatar delivery and animation flow of 3GPP TS 26.264, receiver-centric mode



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IMPLEMENT

Repositories

What each repository covers, from its README; all under the 5G-MAG Public License

REPOSITORY	WHAT IT IS	COVERS	LATEST RELEASE	
rt-avatar-call	the call application: Android and Windows	Receiver-centric call, ARF format, Animation, SWAP signalling	● v0.10.154 16 Oct 2026	GitHub
rt-avatar-content	reference MPEG ARF content for the call	ARF format	● v0.10.154 16 Oct 2026	GitHub
rt-avatar-unity-tracker	a sample OpenXR tracker for ARF avatar animation in Unity; no code yet	Animation	○ none yet	GitHub
rt-avatar-bar	stores base avatar models as ARF containers, scoped to a call; client fetch pending	Base Avatar Repository	○ none yet	GitHub
rt-3gpp-swap	the 3GPP SWAP server and Node.js library (TS 26.113)	SWAP signalling	● v0.9 16 Oct 2026	GitHub

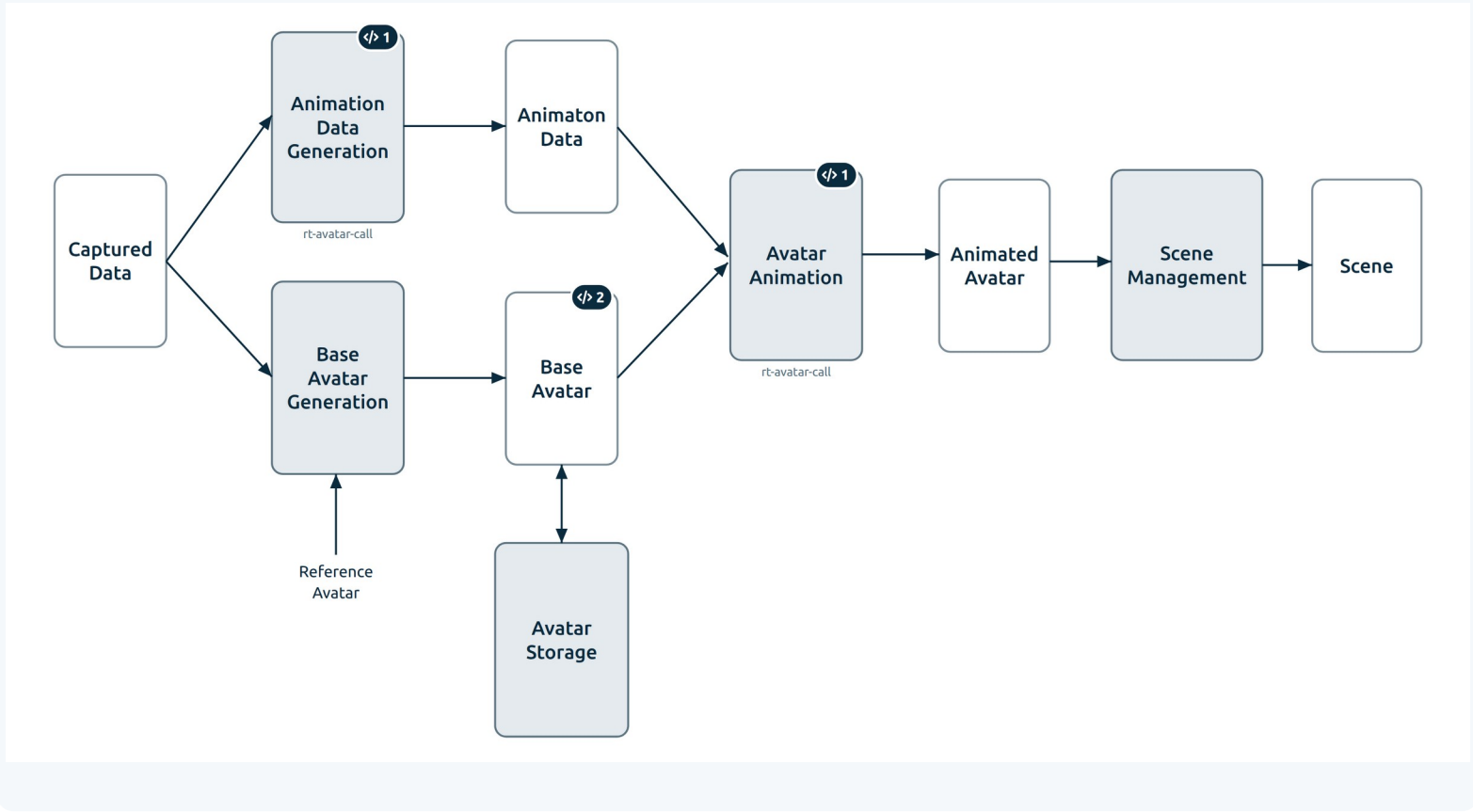
FIND OUT MORE — Technical analysis · Standards · Reference Tools

● released ○ no release yet

More information at 5g-mag.com/tech/avatar

Repositories in the architecture

The Reference Tools on the specification figure, as the figure generator draws them



PROVIDED BY THE REFERENCE TOOLS

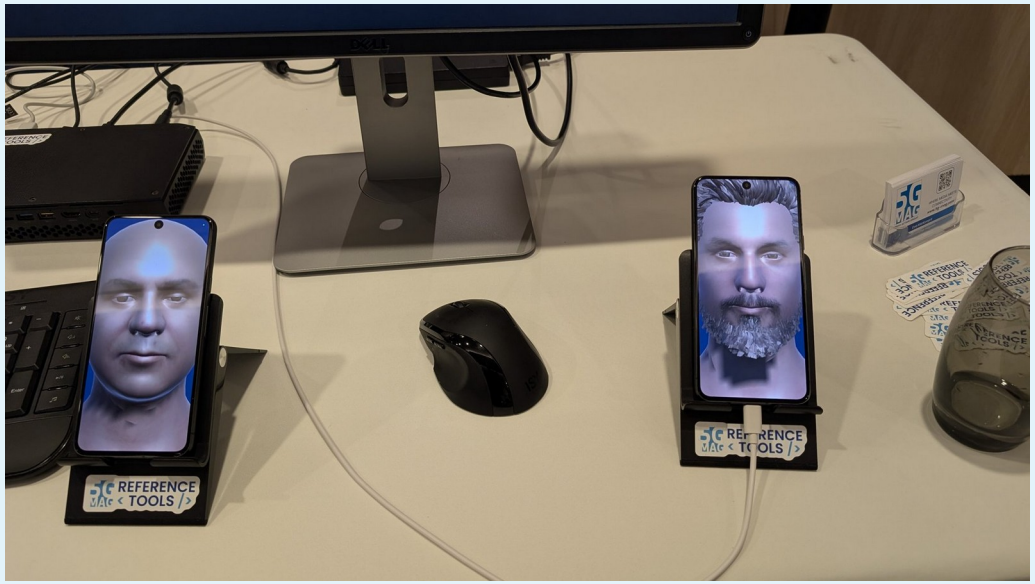
1. **rt-avatar-call**
2. **rt-avatar-content**
3. **MediaPipe**
   dependency, used by 1
4. **UnityARF**
   dependency, used by 1

Boxes without a badge are not mapped to a repository.

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See it and try it

Photos of the project's demo, an avatar call on two phones



MPEG meeting, Geneva, 14 July 2026



IBC 2026, 13 September 2026

CONTRIBUTORS MOTION SPELL Qualcomm interdigital.

CONTRIBUTE

Latest updates and roadmap

Status on 19 October 2026

RELEASED

rt-avatar-call v0.10.154

16 Oct 2026. CLI build tooling and updated default camera framing.

rt-avatar-content v0.10.154

16 Oct 2026. Content supporting rt-avatar-call v0.10.154.

rt-3gpp-swap v0.9

16 Oct 2026. Closes the TS 26.113 SWAP conformance gaps.

IN DEVELOPMENT

v0.10.155

Initial support for Gaussian splatting based avatars, with UnityARF 1.1.0.155; pre-release v0.10.155-b2 is out.

Android XR support

Animation generation on AndroidXR is under development.

Base Avatar Repository

rt-avatar-bar: the service is ready; the client fetch in rt-avatar-call is pending.

PLANNED

Body animation support

Planned for rt-avatar-call.

Call-flow UI

A UI that supports a call flow.

MERGED SINCE V0.9.1 5 pull requests in 3 repositories

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